

Valuation Project 2: Binomial Tree

Choice #4 Lookback option with a simple payoff

Group members: Chia-Chun Hsu, Yu-Fong Lin

Executive Summary

We used the CRR Binomial Tree model to price Lookback Entry Jump Securities with Auto-Callable Feature. Our estimated fair value of the security is **\$980.68** per note.

Parameter Selection

N (Tree steps)	790	r (Risk-free rate)	3.2625%
T (Time)	790 days (1/9/26 ~ 3/14/28)	σ (Implied volatility)	19.146%
S₀ (Initial level)	6,966.28 (Subject to change)	q (Dividend yield)	1.186%
K (Strike level)	100%, 80%	Tree model	CRR model

- **N (Tree steps) - 790**

We chose 790 steps so that each step represents exactly one calendar day. This allows the model to precisely capture the "Lowest Index Level" based on daily closing prices during the 21-day setting period as specified in the term sheet. To ensure the robustness of our model, we performed sensitivity tests (Shown in the latter section) across various step sizes (N=385, 790, 1580). The valuation converged significantly at N=790. This specific choice aligns the tree's temporal grid with the daily closing price observation frequency of the product. Such "Grid Alignment" ensures that the Lookback strike (K) always coincides with the tree nodes, effectively eliminating nonlinearity errors inherent in discrete observation sampling.

By making the number of steps a multiple of the total days, we ensure that critical determination dates—such as the 1/30/26 lookback deadline and the Day 424 Autocall date—fall exactly on tree nodes. This eliminates interpolation errors that occur when key events happen between nodes.

This choice also minimizes "nonlinearity errors" by matching the tree's discrete grid to the product's actual daily observation frequency. It provides a balance between valuation accuracy and the processing time required for a path-dependent algorithm.

- **T (Time) - 790 days (1/9/26 ~ 3/14/28)**

To ensure rigorous valuation, we explicitly defined seven key dates based on the product prospectus.

Following a discussion with the professor, we explicitly set the tree's origin T_0 to 1/9/26 (the inception of the lookback period) to ensure superior numerical integrity.

Elimination of Interpolation Error: If the tree were initiated on 1/16/26, the realized minimum level from 1/9/26~1/15/26 would exist as an external constant. If this constant does not align with the tree's discrete nodes, subsequent calculations for payoff ratios (S/K) and Autocall probabilities would require interpolation, introducing numerical noise.

Unified Discrete Framework: By aligning T_0 with the contract's start date, we ensure that both historical realizations and future projections reside within a unified coordinate system. This structural alignment prevents the leaking of probabilities across misaligned grid boundaries and significantly enhances the precision of valuing path-dependent features.

- T_0 : 1/9/26 Pricing date (The market data is observed and the contract terms are struck.)
- T_1 : 1/22/26 Issue date (The securities are issued and initial funds are exchanged.)
- T_2 : 1/30/26 End of Initial observation period (The initial level will be defined as the lowest closing level from 1/9/26 to 1/30/26.)
- T_3 : 3/9/27 First determination date (The securities will be automatically redeemed if the closing level of the underlier is greater than or equal to the call threshold level)
- T_4 : 3/12/27 Early redemption date (The early redemption payment will be made if the securities have not been automatically redeemed prior to maturity.)
- T_5 : 3/9/28 Final determination date (The underlying asset's final closing price ($S_{\square}$) is recorded to determine the payoff.)
- T_6 : 3/14/28 Maturity date (The final cash flows are actually paid to investors if the securities have not been automatically redeemed prior to maturity.)

Using these dates, we constructed a dual-timeline framework for our Binomial Tree.

- Initial observation Period: Calculated as $T_2 - T_0$ (21 days, or 0.0575 year). The lowest closing level during the initial observation period will be determined as the initial level. In no event will the initial level be greater than: 6,966.28, which is the closing level of the underlier on 1/9/26.
- Automatic early redemption observation Period: Calculated as $T_3 - T_0$ (424 days, or 1.1616 years). The securities will be automatically redeemed if the closing level of the underlier is greater than or equal to the call threshold level on the first determination date for the early redemption payment. No further payments will be made on the securities once they have been automatically redeemed.

- Observation Period: Calculated as $T_5 - T_0$ (790 days, or 2.1644 years). This period dictates the binomial tree and the expected growth of the underlying asset. It begins on the pricing date and ends on the observation date.

- **S_0 (Initial level) - 6,966.28 (Subject to change)**

According to the term sheet, the initial level will be defined as the lowest closing level during the initial observation period (1/9/26 ~ 1/30/26). The closing level on 1/9/26 is 6,966.28.

- **K (Strike level) - 100%, 80%**

For the automatic early redemption, the strike level is 100% of the initial level. The securities will be automatically redeemed if the closing level of the underlier is greater than or equal to the call threshold level on the first determination date for the early redemption payment.

If the securities have not been redeemed, the payment at maturity will be made. If the final level is greater than 100% of the initial level, investors will receive the stated principal (1000) amount plus the upside payment.

If the final level is equal to or less than the initial level but is greater than or equal to the downside threshold level (80% of the initial level), investors will receive only the stated principal amount at maturity.

If, however, the final level is less than the downside threshold level, investors will lose 1% for every 1% decline in the level of the underlier over the term of the securities.

- **r (Risk-free rate) - 3.2625%**

We obtained the risk-free rate and discount factor data from Bloomberg and interpolated the term structure to match the maturity of the security. We searched <USD OIS> on Bloomberg and set the date to 1/7/26 which was two days before the pricing date (1/9/26) due to swap settlement time.

Using OIS for the risk-free rate is correct because it reflects the overnight indexed swap rate, which serves as the modern market standard for a "risk-free" benchmark by minimizing credit and liquidity risk premiums. Since these structured products are often collateralized under Credit Support Annex (CSA) agreements, OIS accurately represents the actual funding and discounting costs used by financial institutions. Incorporating a precise OIS-based discount factor into the binomial tree ensures the valuation model remains consistent with non-arbitrage pricing principles and market hedging practices.

We interpolated the Discount Factors (DF) to ensure smoothness. Using the Bloomberg DF quotes for 1/10/28 (0.936750) and 1/9/29 (0.906261), we performed linear interpolation (59-day out of the 365-day interval) to find the exact DF for our target dates. Subsequently, we reverse-engineered the continuous-compounding risk-free zero rates (r) using the formula: $r = -\ln(\text{DF}) / T$. This methodology yielded a precise zero rate of 3.2625%.

For the smoother implementation, the application of a single, unified risk-free rate within our CRR model is a choice designed to balance valuation accuracy with the computational complexity. We assume a locally flat term structure that remains consistent throughout the life of the security, which isolates the impact of the underlying S&P 500 price dynamics from interest rate fluctuations.

- **σ (Implied volatility) - 19.146%**

Implied volatilities were extracted from Bloomberg <OVM> for the specific moneyness levels (100% and 80%). There are four choices of the implied volatility.

- σ (100%, 21d) = 11.307%: The initial level is determined during this period.
- σ (100%, 424d) = 17.770%: The automatic early redemption is determined during this period.
- σ (100%, 790d) = 19.146%: The final payoff is determined during this period. If the final level is greater than 100% of the initial level, investors will receive the stated principal (1000) amount plus the upside payment.
- σ (80%, 790d) = 23.944%: The final payoff is determined during this period. If the final level is less than the downside threshold level, investors will lose 1% for every 1% decline.

We choose σ (100%, 790d) = 19.146% as our sigma in the model because it most accurately aligns with the terminal payoff structure of the security, which is the primary driver of the instrument's value. Since the product is a lookback option with no interim coupons or call features, the valuation is most sensitive to the index level at the final observation date. While shorter-dated or out-of-the-money volatilities capture specific early-period or downside risks, the 790-day at-the-money volatility provides the most robust estimate.

Furthermore, applying a single, constant volatility parameter is essential for maintaining the recombining property of the CRR binomial tree. This approach prevents the grid from becoming path-dependent in its geometry, thereby reducing computational complexity and ensuring that the model converges efficiently to a stable value

- **q (Dividend yield) - 1.186%**

The dividend yield (1.186%) of the underlying S&P 500 index was obtained from Bloomberg <OVME>. Since the underlying asset is a broad market index, applying a continuous dividend rate is more appropriate and computationally efficient for the binomial tree than modeling discrete dividend payments.

- **Tree model - CRR model**

The CRR (Cox-Ross-Rubinstein) model is the best choice for the lookback options.

$u = \exp(\sigma * \sqrt{dt})$

$d = \exp(-\sigma * \sqrt{dt})$

$$p = (\text{np.exp}((r - q) * dt) - d) / (u - d)$$

- Efficient Recombining Structure: The CRR tree's symmetric property ($u \times d = 1$) ensures that nodes recombine, preventing the exponential growth of the tree and making it computationally feasible to track the path-dependent maximum or minimum index levels.
- Direct Volatility Calibration: Its parameters are explicitly linked to the underlying asset's volatility, which directly addresses the project's requirement to justify parameter choices and perform sensitivity analysis on volatility.
- Stable Convergence: For a standard underlying like the S&P 500, the CRR model provides a more stable and predictable convergence to the continuous-time price as the number of steps increases, helping to minimize the nonlinearity errors highlighted in the assignment.

Algorithm Description

The algorithm implements a three-phase binomial tree to value a path-dependent lookback option. Since the strike price of the security is defined as the lowest index level (Minimum Price) observed during the Initial observation Period (1/9/26 ~ 1/30/26), the model must track historical minimums before transitioning into a standard valuation tree.

Phase 1: Path-Dependent State Tracking (The Lookback Period)

During the first 21 steps (representing the period until 1/30/26), the algorithm cannot use a simple 1D array because the payoff depends on the running minimum of the S&P 500.

$$V_{i,j}(M) = e^{-r \Delta t} [p \cdot V_{i+1,j+1}(\min(M, S_{i+1,j+1})) + (1-p) \cdot V_{i+1,j}(\min(M, S_{i+1,j}))]$$

- State Machine: We use a 2D dictionary-based tree where each node stores a mapping of {possible_minimum_price: option_value}.
- Min-Price Logic: At each step, the algorithm updates the state by comparing the current node's price with the previous minimum ($M_{\text{new}} = \min(M_{\text{old}}, S_{\text{current}})$).
- Backward Induction: We work backward from the end of the setting period, weight the values by risk-neutral probabilities (p and $1-p$), and discount them using the OIS-derived rate.

Phase 2: Forward-Looking Valuation (Post-Setting Period)

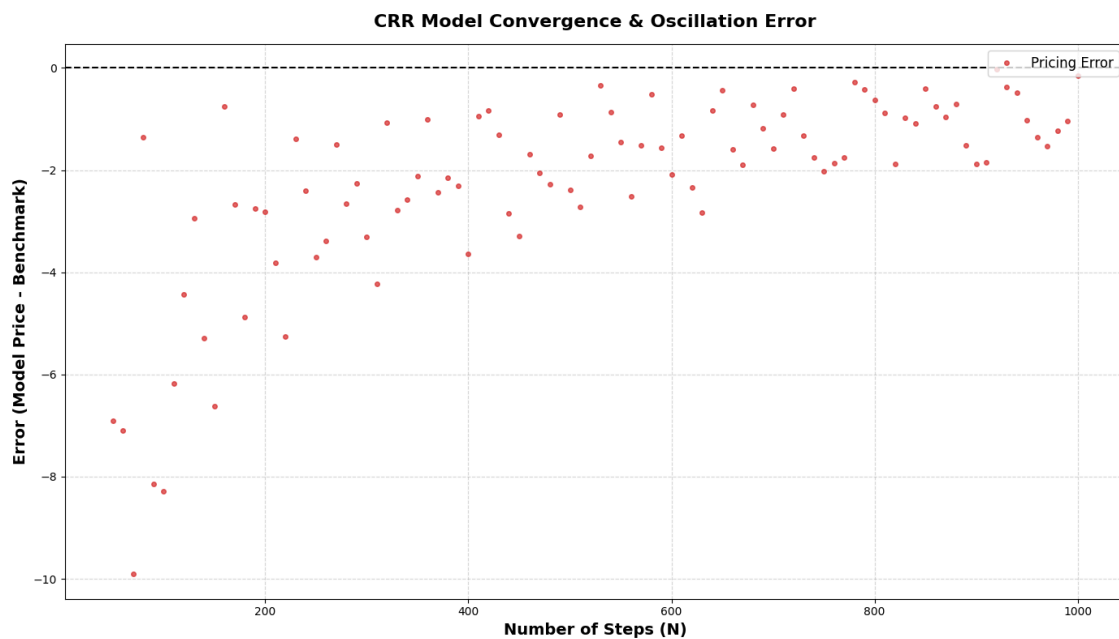
Once the "Lowest Index Level" (K) is fixed on January 30, the product's path-dependency for the strike price ends. The model will be a standard binomial tree with autocall feature.

Phase 3: Final Payoff and Discounting

At maturity, the algorithm calculates the payoff based on three scenarios defined in the term sheet. Then we discount the cashflow from 3/14/26 to 3/9/26.

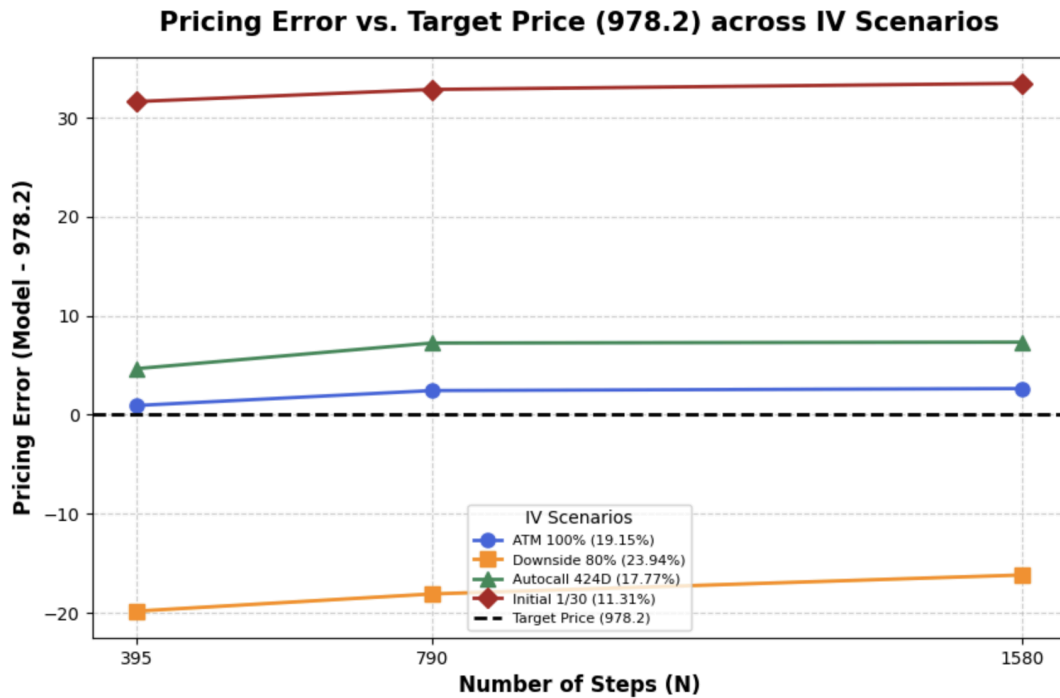
- **Upside Scenario:** If $S_T \geq K$, the investor receives the stated principal amount + (stated principal amount $\times$ participation rate $\times$ underlier percent change (participation rate = 150%).
- **Buffer Scenario:** If S_T is between 80% and 100% of K , the investor receives the \$1000 principal.
- **Downside Scenario:** If $S_T < 80\% K$, the investor loses 1% for every 1% decline below the strike ($\$1,000 \times$ performance factor).

Model robustness test & Sensitivity Analysis



The scatter plot illustrates the **Model Robustness** by analyzing the relationship between the number of steps (N) and the **Convergence Error**. In this analysis, the error is defined as the price difference between each model iteration (from $N=50$ to $N=1,000$) and the benchmark result at $N=2,000$, which serves as our proxy for the "true" numerical value.

- **Convergence Trend:** As N increases from 50 toward 1,000, the calculated price exhibits strong convergence behavior. The narrowing gap toward the $N=2,000$ baseline demonstrates the numerical stability of our two-stage state-machine algorithm as the discrete grid becomes finer.
- **Oscillation Error:** The plot reveals a characteristic "sawtooth" oscillatory pattern, a known numerical artifact of the CRR model. These oscillations occur because the relative position of the lookback strike price (K) within the tree nodes shifts as N varies.
- **Stability at $N=790$:** Our selection of $N=790$ is justified as the model reaches a stable convergence zone where the pricing error relative to the $N=2,000$ benchmark becomes negligible. By comparing against a high-step numerical solution rather than the market price, we effectively isolate the discretization error from other market-driven parameter uncertainties.



IV Scenario	Sigma	Price (395 Steps)	Price (790 Steps)	Price (1580 Steps)
ATM 100% Moneyness	19.146%	979.117	980.627	980.829
Downside 80% Moneyness	23.944%	958.355	960.09	962.008
Autocall 424D Term	17.770%	982.838	985.426	985.517
Initial Set 1/30 Term	11.307%	1009.81	1011.03	1011.65

These two charts demonstrate how our CRR model's valuation deviates from the market benchmark of \$978.20 when applying the different implied volatility levels available from Bloomberg.

Impact of Volatility Choice on Valuation

- **ATM 100% Moneyness (19.146%):** This scenario (blue line) yields the most stable results, with the price at N=790 (\$980.627) tracking closest to the target price. The error remains minimal across all step counts, justifying its use as our primary input.
- **Initial Observation Period (11.307%):** The lower volatility from the 21-day period (red line) significantly overvalues the product (reaching over \$1011). A lower sigma during the period suggests a higher expected "lowest index level" (strike K), which artificially inflates the upside potential.
- **Downside 80% Moneyness (23.944%):** Conversely, using the higher skew volatility (yellow line) results in a substantial underestimation (-\$18 error). The increased volatility raises the probability of the index dropping below the 80% buffer, increasing the expected loss in the downside scenario.

Convergence Stability across N Steps

The line graph compares three step counts: 395, 790, and 1580 (representing 0.5, 1.0, and 2.0 steps per day):

- **Marginal Gains Beyond 790 Steps:** The transition from 790 to 1580 steps shows nearly flat lines for all scenarios. For our chosen ATM IV, the price only moves from \$980.627 to \$980.829.
- **Efficiency of Daily Frequency:** This confirms that our chosen resolution of $N=790$ (one step per day) is the optimal point where the model reaches stability. Increasing the steps further adds computational burden without materially changing the valuation or reducing the error relative to the benchmark.

Possible Error

- **Single Risk-Free Rate Assumption**

While we used a precisely interpolated OIS rate of 3.2625% to match the 790-day maturity, our model assumes a flat term structure throughout the life of the option. In reality, interest rates are stochastic. Using a single rate ignores the drift in the yield curve and the convexity of the discount factors over the 790-day horizon. For a product with no interim coupons, even a slight shift in long-term rates could significantly impact the present value of the final principal repayment.

- **Static Volatility and Surface Risk**

We utilized a single implied volatility of 19.146% (Sigma at 100% moneyness for 790 days) to ensure the binomial tree remains a recombining grid. However, Bloomberg data shows a clear volatility surface where shorter-dated and out-of-the-money options command different premiums. By ignoring the volatility smile, our model may fail to capture the cost of hedging the downside buffer feature (80% threshold) or the skew associated with the S&P 500's potential for sudden drops.

- **Nonlinearity and Discrete Sampling Error**

The product's "Lowest Index Level" is determined by daily closing levels, but a binomial tree is a discrete approximation of a continuous process. Even with $N=790$ steps (one step per day), a nonlinearity error exists because the tree may not perfectly capture the true intra-day minimum or the exact price path between daily nodes. This discrete sampling bias generally leads to an overestimation of the strike price K , which could cause a slight underestimation of the final upside payment compared to a continuous-time model.

Final Recommendation

Our final estimated value is \$980.68 per note by the CRR binomial valuation.

Our analysis confirms that the CRR binomial model with $N=790$ steps provides a robust and computationally efficient valuation for this path-dependent security. We recommend utilizing the ATM 100% Moneyness (19.146%) implied volatility as the primary input, as it minimizes pricing error relative to the market benchmark of \$978.20 and exhibits the most stable

convergence behavior. However, investors should consider the possible errors mentioned in the previous section.